

Recursions

The Geometric Sequence

Definition

A sequence $a_0, a_1, a_2, \dots$ is called a **geometric sequence** if, and only if, there is a constant r such that

$$a_k = ra_{k-1} \quad \text{for each integer } k \geq 1.$$

It follows that

$$a_n = a_0 r^n \quad \text{for each integer } n \geq 0.$$

From one-order to second-order

Definition

A **second-order linear homogeneous recurrence relation with constant coefficients** is a recurrence relation of the form

$$a_k = Aa_{k-1} + Ba_{k-2} \quad \text{for every integer } k \geq \text{some fixed integer,}$$

where A and B are fixed real numbers with $B \neq 0$.

Suppose that for some number t with $t \neq 0$, the sequence

$$1, t, t^2, t^3, \dots, t^n, \dots$$

satisfies the relation.

This means that each term of the sequence equals A times the previous term plus B times the term before that. So for each integer $k \geq 2$, $t^k = At^{k-1} + Bt^{k-2}$.

From one-order to second-order

continued

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$$a_k = Aa_{k-1} + Ba_{k-2} \quad \text{for every integer } k \geq \text{some fixed integer,}$$

where A and B are fixed real numbers with $B \neq 0$.

$$t^k = At^{k-1} + Bt^{k-2}.$$

In particular, when $k = 2$, the equation becomes

$$t^2 = At + B, \quad \Rightarrow \quad t^2 - At - B = 0.$$



characteristic equation

The Distinct-Roots Case

$$t^2 - At - B = 0.$$

Suppose $r_0, r_1, r_2, \dots$ and $s_0, s_1, s_2, \dots$ are sequences that satisfy the same second-order linear homogeneous recurrence relation with constant coefficients. In other words, suppose that for some real numbers A and B :

$$r_k = Ar_{k-1} + Br_{k-2} \quad \text{and} \quad s_k = As_{k-1} + Bs_{k-2}$$

Lemma 5.8.2

If $r_0, r_1, r_2, \dots$ and $s_0, s_1, s_2, \dots$ are sequences that satisfy the same second-order linear homogeneous recurrence relation with constant coefficients, and if C and D are *any* numbers, then the sequence $a_0, a_1, a_2, \dots$ defined by the formula

$$a_n = Cr_n + Ds_n \quad \text{for every integer } n \geq 0$$

also satisfies the same recurrence relation.

The Distinct-Roots Case

continued

Lemma 5.8.2

If $r_0, r_1, r_2, \dots$ and $s_0, s_1, s_2, \dots$ are sequences that satisfy the same second-order linear homogeneous recurrence relation with constant coefficients, and if C and D are *any* numbers, then the sequence $a_0, a_1, a_2, \dots$ defined by the formula

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also satisfies the same recurrence relation.

The Single-Root Case

$$t^2 - At - B = 0.$$

One sequence that satisfies the recurrence relation is

$$1, r, r^2, r^3, \dots, r^n, \dots$$

But another sequence that also satisfies the relation is

$$0, r, 2r^2, 3r^3, \dots, nr^n, \dots$$

To see why this is so, observe that since r is the unique root of $t^2 - At - B = 0$, the left-hand side of the equation can be written in the form $(t - r)^2$, and so

$$t^2 - At - B = (t - r)^2 = t^2 - 2rt + r^2.$$

$$\Rightarrow \quad A = 2r \quad \text{and} \quad B = -r^2.$$

Let $s_0, s_1, s_2, \dots$ be the sequence defined by the formula

$$s_n = nr^n \quad \text{for each integer } n \geq 0.$$

The Single-Root Case

continued

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$$s_n = nr^n \quad \text{for each integer } n \geq 0.$$

$$A = 2r \quad \text{and} \quad B = -r^2.$$

$$\begin{aligned} As_{k-1} + Bs_{k-2} &= A(k-1)r^{k-1} + B(k-2)r^{k-2} \\ &= 2r(k-1)r^{k-1} - r^2(k-2)r^{k-2} \\ &= 2(k-1)r^k - (k-2)r^k \\ &= (2k-2-k+2)r^k \\ &= kr^k \\ &= s_k \end{aligned}$$

The Single-Root Case

continued

Theorem 5.8.5 Single-Root Theorem

Suppose a sequence $a_0, a_1, a_2, \dots$ satisfies a recurrence relation

$$a_k = Aa_{k-1} + Ba_{k-2}$$

for some real numbers A and B with $B \neq 0$ and for every integer $k \geq 2$. If the characteristic equation $t^2 - At - B = 0$ has a single (real) root r , then $a_0, a_1, a_2, \dots$ is given by the explicit formula

$$a_n = \underline{Cr^n} + \underline{Dnr^n},$$

where C and D are the real numbers whose values are determined by the values of a_0 and any other known value of the sequence.

The Single-Root Case

Exercise: Given recurrence relation $a_n = 2a_{n-1} - a_{n-2}$, $a_0 = C, a_1 = C + D$. Please use mathematical induction to prove that $a_n = C + Dn$.

The Single-Root Case

continued

Exercise: Given recurrence relation $a_n = 2a_{n-1} - a_{n-2}$, $a_0 = C$, $a_1 = C + D$. Please use mathematical induction to prove that $a_n = C + Dn$.

Solution:

Base cases: verify the solution for $n = 0$ and $n = 1$:

$$a_0 = C + D * 0 = C$$

$$a_1 = C + D * 1 = C + D$$

Inductive hypothesis: assume the solution holds for $n = k$ and $n = k - 1$, i.e.,

$$a_k = C + Dk \quad \text{and} \quad a_{k-1} = C + D(k - 1)$$

The Single-Root Case

continued

Exercise: Given recurrence relation $a_n = 2a_{n-1} - a_{n-2}$, $a_0 = C$, $a_1 = C + D$. Please use mathematical induction to prove that $a_n = C + Dn$.

Solution:

Goal: show it holds for $n = k + 1$, i.e.,

$$a_{k+1} = C + D(k + 1)$$

Proof:

Using the recurrence relation $a_{k+1} = 2a_k - a_{k-1}$

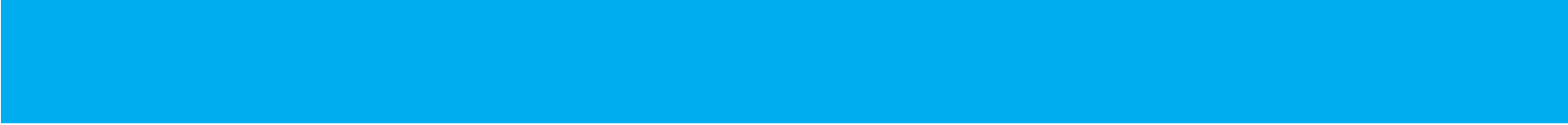
Substitute the inductive hypothesis:

$$a_{k+1} = 2(C + Dk) - (C + D(k - 1)) = C + D(k + 1)$$

Conclusion: By mathematical induction, the solution $a_n = C + Dn$ satisfies the recurrence relation for all $n \geq 0$

From Second-Order to Any-Order?

$$a_n = Aa_{n-1} + Ba_{n-2} + Ca_{n-3} + \cdots$$



Correctness of the Division Algorithm

Correctness of the Division Algorithm

The division algorithm is supposed to take a nonnegative integer a and a positive integer d and compute nonnegative integers q and r such that $a = dq + r$ and $0 \leq r < d$.

Initially, the variables r and q are introduced and given the values $r = a$ and $q = 0$.

Correctness of the Division Algorithm

The crucial loop, annotated with pre- and post-conditions, is the following:

*[Pre-condition: a is a nonnegative integer
and d is a positive integer, $r = a$, and $q = 0$.]*

while ($r \geq d$)

1. $r := r - d$

2. $q := q + 1$

end while

*[Post-condition: q and r are nonnegative integers
with the property that $a = qd + r$ and $0 \leq r < d$.]*

Correctness of the Division Algorithm

continued

*[Pre-condition: a is a nonnegative integer
and d is a positive integer, $r = a$, and $q = 0$.]*

while ($r \geq d$)

1. $r := r - d$

2. $q := q + 1$

end while

Loop invariante:

$I(n): r = a - nd \geq 0 \quad \text{and} \quad n = q.$

Guard: $r \geq d$

*[Post-condition: q and r are nonnegative integers
with the property that $a = qd + r$ and $0 \leq r < d$.]*

I. Basis Property: *[$I(0)$ is true before the first iteration of the loop.]*

Correctness of the Division Algorithm

continued

*[Pre-condition: a is a nonnegative integer
and d is a positive integer, $r = a$, and $q = 0$.]*

while $(r \geq d)$

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end while

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II. Inductive Property: *[If $G \wedge I(k)$ is true before an iteration
of the loop (where $k \geq 0$), then $I(k+1)$ is true after iteration
of the loop.]*

Correctness of the Division Algorithm

continued

*[Pre-condition: a is a nonnegative integer
and d is a positive integer, $r = a$, and $q = 0$.]*

while ($r \geq d$)

1. $r := r - d$

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Loop invariante:

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Guard: $r \geq d$

*[Post-condition: q and r are nonnegative integers
with the property that $a = qd + r$ and $0 \leq r < d$.]*

III. Eventual Falsity of the Guard: *[After a finite number of iterations of the loop, G becomes false.]*

Correctness of the Division Algorithm

continued

*[Pre-condition: a is a nonnegative integer
and d is a positive integer, $r = a$, and $q = 0$.]*

while ($r \geq d$)

1. $r := r - d$

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end while

Loop invariante:

$I(n): r = a - nd \geq 0 \quad \text{and} \quad n = q.$

Guard: $r \geq d$

*[Post-condition: q and r are nonnegative integers
with the property that $a = qd + r$ and $0 \leq r < d$.]*

IV. Correctness of the Post-Condition: *[If N is the least number of iterations after which G is false and $I(N)$ is true, then the values of the algorithm variables will be as specified in the post-condition of the loop.]*

Example 5.6.6 – *The Fibonacci Numbers*

One of the earliest examples of a recursively defined sequence occurs in the writings of Leonardo of Pisa, commonly known as Fibonacci, who was the greatest European mathematician of the Middle Ages and promoted the use of Hindu-Arabic numerals for calculation. In 1202 Fibonacci posed the following problem:

A single pair of rabbits (male and female) is born at the beginning of a year.

Example 5.6.6 – *The Fibonacci Numbers*_{continued}

Assume the following conditions:

1. Rabbit pairs are not fertile during their first month of life but thereafter give birth to one new male/female pair at the end of every month.
2. No rabbits die.

How many rabbits will there be at the end of the year?

Example 5.6.6 – *Solution*

One way to solve this problem is to plunge right into the middle of it using recursion. Suppose you know how many rabbit pairs there were at the ends of previous months. How many will there be at the end of the current month?

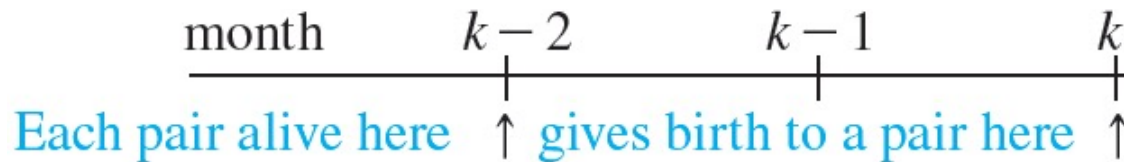
The crucial observation is that the number of rabbit pairs born at the end of month k is the same as the number of pairs alive at the end of month $k - 2$. Why?

Because it is exactly the rabbit pairs that were alive at the end of month $k - 2$ that were fertile during month k .

Example 5.6.6 – Solution

continued

The rabbits born at the end of month $k - 1$ were not.



Now the number of rabbit pairs alive at the end of month k equals the ones alive at the end of month $k - 1$ plus the pairs newly born at the end of the month. Thus

$$\left[\begin{array}{l} \text{the number} \\ \text{of rabbit} \\ \text{pairs alive} \\ \text{at the end} \\ \text{of month } k \end{array} \right] = \left[\begin{array}{l} \text{the number} \\ \text{of rabbit} \\ \text{pairs alive} \\ \text{at the end} \\ \text{of month } k - 1 \end{array} \right] + \left[\begin{array}{l} \text{the number} \\ \text{of rabbit} \\ \text{pairs born} \\ \text{at the end} \\ \text{of month } k \end{array} \right]$$

Example 5.6.6 – *Solution*

continued

$$= \left[\begin{array}{l} \text{the number} \\ \text{of rabbit} \\ \text{pairs alive} \\ \text{at the end} \\ \text{of month } k-1 \end{array} \right] + \left[\begin{array}{l} \text{the number} \\ \text{of rabbit} \\ \text{pairs alive} \\ \text{at the end} \\ \text{of month } k-2 \end{array} \right]. \quad 5.6.2$$

For each integer $n \geq 1$, let

$$F_n = \left[\begin{array}{l} \text{the number of rabbit pairs} \\ \text{alive at the end of month } n \end{array} \right]$$

and let

$$\begin{aligned} F_0 &= \text{the initial number of rabbit pairs} \\ &= 1. \end{aligned}$$

Example 5.6.6 – *Solution*

continued

Then by substitution into equation (5.6.2), for every integer $k \geq 2$,

$$F_k = F_{k-1} + F_{k-2}.$$

Now $F_0 = 1$, as already noted, and $F_1 = 1$ also, because the first pair of rabbits is not fertile until the second month.

Hence the complete specification of the Fibonacci sequence is as follows: For every integer $k \geq 2$,

(1) $F_k = F_{k-1} + F_{k-2}$ recurrence relation

(2) $F_0 = 1, \quad F_1 = 1$ initial conditions.

Example 5.6.6 – *Solution*

continued

To answer Fibonacci's question, compute F_2 , F_3 , and so forth through F_{12} :

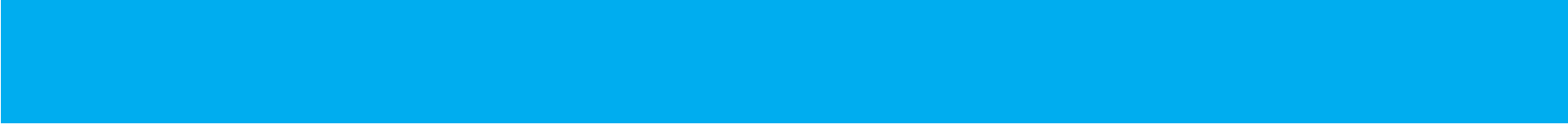
- (3) $F_2 = F_1 + F_0 = 1 + 1 = 2$ by (1) and (2)
- (4) $F_3 = F_2 + F_1 = 2 + 1 = 3$ by (1), (2), and (3)
- (5) $F_4 = F_3 + F_2 = 3 + 2 = 5$ by (1), (3), and (4)
- (6) $F_5 = F_4 + F_3 = 5 + 3 = 8$ by (1), (4), and (5)
- (7) $F_6 = F_5 + F_4 = 8 + 5 = 13$ by (1), (5), and (6)
- (8) $F_7 = F_6 + F_5 = 13 + 8 = 21$ by (1), (6), and (7)

Example 5.6.6 – *Solution*

continued

(9)	$F_8 = F_7 + F_6 = 21 + 13$	$= 34$	by (1), (7), and (8)
(10)	$F_9 = F_8 + F_7 = 34 + 21$	$= 55$	by (1), (8), and (9)
(11)	$F_{10} = F_9 + F_8 = 55 + 34$	$= 89$	by (1), (9), and (10)
(12)	$F_{11} = F_{10} + F_9 = 89 + 55$	$= 144$	by (1), (10), and (11)
(13)	$F_{12} = F_{11} + F_{10} = 144 + 89$	$= 233$	by (1), (11), and (12)

At the end of the twelfth month there are 233 rabbit pairs, or 466 rabbits in all.



Proving Properties about Recursively Defined Sets

Proving Properties about Recursively Defined Sets

Theorem 5.9.2 Additive Property of String Length

If S is the set of all strings over a finite set A , then for all strings u and v in S , $L(uv) = L(u) + L(v)$.

Proof (by structural induction): Let S be the set of all strings over a finite set A . Given any string v in S , let the property $P(v)$ be the sentence

$$\text{For every string } u \text{ in } S, \quad L(uv) = L(u) + L(v).$$

We will show that $P(v)$ is true for every string v in S .

Show that $P(a)$ is true for each string a in the base for S :

The only string in the base for S is λ , and if u is any string in S , then

$$\begin{aligned} L(u\lambda) &= L(u) && \text{by part II(b) in the definition of string} \\ &= L(u) + 0 && \text{by definition of } L. \\ &= L(u) + L(\lambda) \end{aligned}$$

This shows that $P(\lambda)$ is true.

Proving Properties about Recursively Defined Sets continued

Theorem 5.9.2 Additive Property of String Length

If S is the set of all strings over a finite set A , then for all strings u and v in S , $L(uv) = L(u) + L(v)$.

Show that for each string x in S , if $P(x)$ is true and if y is obtained from x by applying a rule from the recursion for S , then $P(y)$ is true:

The recursion for S consists of three rules denoted II(a), II(b), and II(c), but rule II(a) is the only one that generates new strings in S . Suppose v is any string in S such that $P(v)$ is true. In other words, suppose that $L(uv) = L(u) + L(v)$. [This is the inductive hypothesis.]

When rule II(a) is applied to v , the result is vc , where c is a character in A . So, to complete the inductive step, we must show that $P(vc)$ is true. Now

$L(u(vc)) = L((uv)c)$	by part II(c) of the definition of string
$= L(uv) + 1$	by definition of length of a string
$= (L(u) + L(v)) + 1$	because u is assumed to satisfy property P
$= L(u) + (L(v) + 1)$	by the associative law for addition
$= L(u) + L(vc)$	by definition of length of a string.

Proving Properties about Recursively Defined Sets

Theorem 5.9.4 Concatenation of Strings Is Associative

If S is the set of all strings over a finite set A and u , v , and w are any strings in S , then $u(vw) = (uv)w$.

$P(w)$: For all strings u and v in S , $u(vw) = (uv)w$.

Base case: $P(\lambda)$: $u(v\lambda) = (uv)\lambda$

Inductive hypothesis: if w is any strings in S such that $P(w)$ is true: $u(vw) = (uv)w$. If y is obtained from w by applying a rule from the recursion for S , we must show that $P(y) = P(wc)$ is true: $u(v(wc)) = (uv)(wc)$

Proof:

$u(v(wc)) = u((vw)c)$	by part II(c) of the definition of string
$= (u(vw))c$	by part II(c) of the definition of string
$= ((uv)w)c$	by $P(w)$
$= (uv)(wc)$	by part II(c) of the definition of string